The Main Monetary Policy Shock*

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Abstract

We propose a novel procedure to identify a main monetary policy (MMP) shock as the linear combination of monetary policy surprises that best explains macroeconomic fluctuations. Implemented with two standard high-frequency yield-curve surprises and the Boehm and Kroner (2024) non-yield shock, the MMP shock explains nontrivial business-cycle variation over the last 30 years, in contrast with prior findings. Moreover, the MMP shock differs markedly from a textbook monetary policy shock: a contractionary shock leaves interest rates mostly unchanged but persistently lowers headline prices and appreciates the dollar, while temporarily tightening financial conditions and depressing industrial production.

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1 Introduction

Conventional macroeconomic wisdom holds that monetary policy shocks have not been important drivers of the U.S. business cycle in recent decades. This view is in part based on empirical evidence and in part on the consensus that monetary policy has been conducted so systematically over the past several decades that it is simply not plausible that the unsystematic "shock" component is an important driving force of the business cycle (e.g., Leeper, Sims, and Zha, 1996). In her review of the literature, Ramey (2016, p. 111) concludes that "it is likely that monetary shocks are no longer an important source of macro instability." Similarly, Plagborg-Møller and Wolf (2022, p. 2167) find that "monetary shocks are almost irrelevant for aggregate inflation in [the] post-1990 sample."

These conclusions, however, rest on the premise that monetary policy news can be summarized in a single dimension—typically, a surprise in the policy rate or its near-term path. By contrast, a sizable event-study literature documents that over narrow windows around FOMC announcements, monetary policy news is multidimensional. Gürkaynak, Sack, and Swanson (2005) show that, in addition to surprises in the near-term policy rate ("target" surprises), surprises about longer-term rates ("forward guidance" surprises) are important. Swanson (2021) extends this decomposition to three factors, interpreted as target, forward guidance, and asset purchase surprises. More recently, Boehm and Kroner (2024) show that monetary policy affects stock prices and exchange rates through channels not spanned by the yield curve and estimate a "non-yield" monetary policy shock. Taken together, this evidence suggests that over narrow event windows at least four dimensions of monetary policy shocks are detectable in the data.[1]

In this paper, we account for this multidimensionality and show that monetary policy shocks both have stronger effects on macroeconomic variables and are more important for U.S. business-cycle fluctuations than previously thought. We begin by proposing a new method to study the effects of multiple monetary policy surprises. Building on Faust (1998), Barsky and Sims (2011), and others, we construct a main monetary policy (MMP) shock as a linear combination of high-frequency monetary policy surprises included in a vector autoregression (VAR). This linear combination is chosen to maximize its contribution to the forecast error variances in real activity and inflation. Our analysis of the resulting MMP shock yields two main findings. First, monetary policy shocks explain substantially more variation than in most comparable studies, particularly for aggregate inflation and the dollar

[1] See also Lewis (2025) and Cieslak and Khorrami (2026), among others, on the multidimensionality of monetary policy news.

exchange rate. Second, the main discrepancy with prior work arises because we include the non-yield shock of Boehm and Kroner (2024) in the VAR, which has large effects on several macroeconomic variables.

To construct the MMP shock, we estimate a VAR in which the monetary block includes (i) a monetary policy surprise about near-term rates (similar to Bauer and Swanson, 2023b), (ii) a surprise about the path of interest rates between two and 30 years ahead (similar to Gürkaynak, Sack, and Swanson, 2005; Swanson, 2021), and (iii) the non-yield shock of Boehm and Kroner (2024). An MMP shock constructed from these three surprises differs markedly from a textbook monetary policy shock, in part because it leaves interest rates mostly unchanged. In the short run, it resembles a demand shock. A contractionary realization lowers industrial production and the consumer price index (CPI), tightens financial conditions, and appreciates the dollar. While most of its effects fully revert after two years, the effects on the CPI and the dollar are large and persistent. On average over four years, the MMP shock explains about 9 percent of the variation in industrial production, about 18 percent of the variation in the CPI, and about 16 percent of the variation in the broad dollar index.

Our analysis highlights a missing element in empirical studies of the macroeconomic effects of monetary policy shocks. After providing evidence for the existence of the non-yield (NY) shock, Boehm and Kroner (2024) show that it has large and significant effects on a range of asset prices at high frequencies, including stocks, exchange rates, and commodity prices. They also show that the NY shock arises in the data whenever monetary policy news is sufficiently high-dimensional that its effects are not fully spanned by the yield curve. Examples include certain policy promises, information effects as in Jarociński and Karadi (2020), and many more. This paper adds to these findings by demonstrating that the NY shock also has important effects on macroeconomic and financial variables at business cycle frequencies.

Our results also point to potentially underappreciated channels of the monetary transmission mechanism. As discussed above, the MMP shock has large effects on headline inflation and the dollar exchange rate. Further analysis reveals that its effect on headline inflation is predominantly driven by a large change in oil prices, while the core CPI is only mildly affected. This suggests that monetary policy may have substantial effects on the U.S. economy through its impact on commodity prices. We further document that the strong dollar appreciation following a contractionary MMP shock is accompanied by a significant decline in U.S. net exports. Given the overall importance of commodity prices and the dollar for

the international transmission of monetary policy (e.g., Degasperi, Hong, and Ricco, 2026), our findings suggest that global spillovers of U.S. monetary policy may be stronger than previously found.

Our paper contributes to the large literature on the effects of monetary policy shocks on the U.S. and the global economy, including Gertler and Karadi (2015); Miranda-Agrippino and Rey (2020); Miranda-Agrippino and Ricco (2021); Bauer and Swanson (2023b); Degasperi, Hong, and Ricco (2026); Aruoba and Drechsel (Forthcoming) and many more. A smaller set of papers considers multiple dimensions of monetary policy news (e.g., Lakdawala, 2019; Jarociński and Karadi, 2020; Acosta, 2022; Swanson, 2023; Jarociński and Karadi, 2026), but none of these papers includes the NY shock. Closest to our paper is Bauer, Schmeling, and Schrimpf (2026, BSS), who also report variance decompositions for multiple monetary policy shocks and whose work we discuss below.

Our econometric approach differs from existing ones in two ways. First, we do not use the proxy-VAR approach that has become standard since Gertler and Karadi (2015). Instead of using high-frequency surprises as instruments, we include them in the VAR and treat their innovations as shocks, an approach that does not require invertibility (Plagborg-Møller and Wolf, 2021) and avoids the scaling inherent in IV estimation. As a result, our estimates are not subject to the weak-instrument problems that often afflict proxy VARs (Lewis and Mertens, 2026), and we can combine several surprises with limited variation. The main cost is that our variance shares are lower bounds: because the surprises miss monetary news outside FOMC announcement windows, such as speeches by Fed officials (Swanson and Jayawickrema, 2024), treating them as shocks understates the importance of monetary policy (Plagborg-Møller and Wolf, 2022).

Second, we do not take a stance on how to identify individual structural monetary policy shocks. Such identification schemes (or rotations) have proved controversial; for instance, the importance of information effects remains disputed (Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Bauer and Swanson, 2023a, and others). Boehm and Kroner (2024) further show that the existence of the NY shock generally complicates the identification of structural monetary policy shocks because it arises under very different sets of assumptions. Our estimation procedure avoids taking a stance on the correct rotation by asking a different research question: Our goal is not to identify the causal effects of particular *structural* monetary policy shocks. Instead, our goal is to identify the linear combination of them that best explains certain outcome variables. Under our identification assumption—that the monetary surprises are exclusively spanned by structural monetary policy shocks—

this approach can be implemented using only the VAR residuals of the monetary policy surprises, that is, without identifying the structural monetary policy shocks. An added benefit of our approach is that it returns a time series of the MMP shock, capturing the most important variation of U.S. monetary policy disturbances for the U.S. business cycle.

Both differences matter when comparing our results with the closely related work of BSS. BSS use a high-frequency risk appetite surprise, positively correlated with the NY shock (0.45), and an interest-rate surprise as external instruments in a proxy SVAR. Like us, they find that monetary policy news not captured by interest-rate surprises has large macroeconomic effects. Unlike us, they impose an additional restriction to separate an interest-rate shock from a risk appetite shock, which lets them speak more directly to transmission channels. Their instruments are weak (first-stage F-statistics of about 2); BSS argue this mainly reduces precision, but Lewis and Mertens (2026) show that weak-instrument bias remains a concern in such environments. Finally, their variance shares are larger than ours, especially for consumer prices, partly for methodological reasons: in the framework of Plagborg-Møller and Wolf (2022), treating surprises as shocks—as we do—yields a lower bound on a shock's variance contribution, whereas proxy SVARs impose invertibility, under which an upper bound of its variance contribution is obtained.

Roadmap The remainder of the paper is structured as follows. The next section provides a brief review of the Boehm and Kroner (2024) non-yield shock. Section 3 then lays out the general procedure for estimating the MMP shock. In Section 4 we estimate a specific version of the MMP shock using two yield surprises and the NY shock and present the results. Section 5 concludes.

2 The Fed Non-yield Shock

Boehm and Kroner (2024, henceforth BK) start from the following fact: high-frequency monetary policy surprises à la Kuttner (2001), Gürkaynak, Sack, and Swanson (2005), and Swanson (2021) explain little of the movements in stock prices and exchange rates in windows around FOMC announcements. This is puzzling, since one would expect the announcement to be by far the most important event within these windows. BK exploit this unexplained variation to construct the NY shock. Roughly speaking, they regress the changes in the S&P 500 and dollar exchange rates around each announcement on changes in the entire yield curve and then extract the NY shock as a common factor in the residuals. By construction, the residuals, and therefore the NY shock, are orthogonal to all yield changes over the event window. This orthogonality is a statistical feature of the construction, not an assumption

about how monetary policy works.

A negative NY shock lowers stock prices and appreciates the dollar while leaving the yield curve unchanged; we refer to it as a contractionary NY shock in this paper. Three findings imply that the NY shock reflects monetary policy. First, for all assets, residual variances are significantly higher on FOMC days than at similar times on other days. Second, correlations between stock prices and exchange rates (and among exchange rates) also rise on FOMC days, suggesting that a common shock drives the co-movement of these asset prices. Third, market commentary on days with large NY shocks attributes the market movements to the Fed. Taken together, the NY shock represents a dimension of monetary policy that is not captured by the yield curve.

BK show that the NY shock exists whenever there are more structural monetary policy shocks than factors detectable in the yield curve. One possibility is that the NY shock is a single structural shock that does not move interest rates, such as certain news about the systematic conduct of policy. For instance, a credible promise to ease more aggressively in adverse states lowers expected rates in those states but also makes them less likely. In a knife-edge case, expected rates remain unchanged, while stock prices rise as downside risk falls. However, this is a special case: BK show that the NY shock is structural only if one of the structural monetary policy shocks leaves the entire yield curve unchanged. In general, the NY shock is a reduced-form shock—that is, a linear combination of structural shocks—and its existence does not imply that the underlying structural shocks leave interest rates unchanged.[2] BK's narrative analysis suggests that the NY shock reflects indeed different types of policy news across announcements rather than a single structural shock. For our purposes, the precise interpretation is immaterial: either way, the NY shock is driven only by structural monetary policy shocks, which is what our identification procedure we apply below requires.

3 Empirical Strategy

We identify an MMP shock as the linear combination of monetary policy surprises that maximizes the explained variation of a weighted average of macroeconomic variables included in the VAR. This empirical strategy builds on the maximum forecast-error-variance approach of Faust (1998), Barsky and Sims (2011) and others.

[2] In the framework of Jarociński and Karadi (2020), for example, a contractionary "pure" monetary policy shock raises interest rates and lowers stock prices, whereas a negative information shock lowers both. Combined in the right proportion, the two leave interest rates unchanged but unambiguously lower stock prices.

3.1 Setup

Consider a length-k vector Y_t of monetary policy shocks, macroeconomic variables, and financial variables. Partition this vector into a vector of monetary policy shocks M_t with k_M elements and a vector of macroeconomic and financial variables X_t with $k_X = k - k_M$ elements:

$$Y_t = \begin{pmatrix} M_t \\ X_t \end{pmatrix}.$$

For this identification approach to yield a nontrivial MMP shock, we require that M_t includes at least two different monetary policy surprises, that is, $k_M \geq 2$.

We begin with estimating the following reduced-form VAR with p lags:

$$Y_t = \sum_{j=1}^{p} B_j Y_{t-j} + u_t, \tag{1}$$

where u_t denotes the vector of reduced-form innovations and $E\left[u_t u_t'\right] = \Sigma$. The corresponding moving-average representation is

$$Y_t = \sum_{\tau=0}^{\infty} \Phi_\tau u_{t-\tau}, \tag{2}$$

where Φ_τ denotes the moving-average coefficient matrix, satisfying $\Phi_0 = I$ and $\Phi_\tau = \sum_{j=1}^{\min(\tau,p)} B_j \Phi_{\tau-j}$.

3.2 Identification of the Main Monetary Policy Shock

We posit that the MMP shock is a linear combination of the reduced-form innovations associated with the monetary policy surprises in M_t:

$$\text{MMPS}_t = \alpha' u_t \qquad \text{where} \qquad \alpha = \begin{pmatrix} \alpha_M \\ 0_{k_X \times 1} \end{pmatrix}. \tag{3}$$

The zero restrictions on the last k_X elements of α ensure that the MMP shock is constructed exclusively from the reduced-form innovations of monetary policy surprises. We normalize the MMP shock to have unit variance,

$$\text{Var}\left[\text{MMPS}_t\right] = \alpha' \Sigma \alpha = 1. \tag{4}$$

To understand how the MMP shock depends on the underlying structural shocks ε_t,

consider the mapping $u_t = S\varepsilon_t$ and partition the vectors and matrices as follows:

$$u_t = \begin{pmatrix} u_t^M \\ u_t^X \end{pmatrix}, \qquad S = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix}, \qquad \varepsilon_t = \begin{pmatrix} \varepsilon_t^M \\ \varepsilon_t^X \end{pmatrix}.$$

The MMP shock can then be written as

$$\text{MMPS}_t = \alpha' S\varepsilon_t = \alpha_M' \left(S_{11}\varepsilon_t^M + S_{12}\varepsilon_t^X \right).$$

This expression highlights the main identification assumption we require: $S_{12} = 0$. In words, we assume that the structural shocks from the macro-finance block of the VAR, ε_t^X, do not contemporaneously impact the monetary policy surprises included in the VAR. High-frequency identified monetary surprises are likely to satisfy this assumption, at least after an appropriate purification (Bauer and Swanson, 2023a,b). Under this assumption, the MMP shock captures a linear combination of *structural monetary* policy shocks only. Since our estimation procedure will not identify the matrix S_{11}, but only the vector α_M, the dependence of the MMP shock on the underlying structural shocks will remain unknown.

Next, consider the linear projection of the VAR residuals onto the MMP shock

$$u_t = b \cdot \text{MMPS}_t + v_t. \tag{5}$$

The coefficient b is determined by ordinary least squares (OLS):

$$b = \frac{\Sigma\alpha}{\alpha'\Sigma\alpha} = \Sigma\alpha. \tag{6}$$

As just noted, under the assumption that $S_{12} = 0$ the MMP shock is only a function of structural monetary policy shocks. Equation (5) is therefore not subject to reverse causality concerns. Instead, the estimation of b by OLS is part of our statistical procedure to identify the MMP shock.

The forecast error of Y_{t+h} is

$$Y_{t+h} - E_{t-1}\left[Y_{t+h}\right] = \sum_{\tau=0}^{h} \Phi_\tau u_{t+h-\tau}. \tag{7}$$

Letting e_i denote a selection vector, the forecast-error variance of variable i is then

$$FEV_i(h) = e_i' \left(\sum_{\tau=0}^{h} \Phi_\tau \Sigma \Phi_\tau' \right) e_i. \tag{8}$$

Using the predicted value from regression (5) in equation (7) and taking the variance shows that the share of forecast-error variance of variable i attributable to the MMP shock is

$$\Omega_{i,\text{MMPS}}(h) = \frac{\sum_{\tau=0}^{h} e_i' \Phi_\tau b \sigma^2_{\text{MMPS}} b' \Phi_\tau' e_i}{e_i' \left(\sum_{\tau=0}^{h} \Phi_\tau \Sigma \Phi_\tau' \right) e_i} = \frac{\sum_{\tau=0}^{h} e_i' \Phi_\tau \Sigma \alpha \alpha' \Sigma \Phi_\tau' e_i}{e_i' \left(\sum_{\tau=0}^{h} \Phi_\tau \Sigma \Phi_\tau' \right) e_i}. \tag{9}$$

where the second equality uses (6) and the fact that $\sigma^2_{\text{MMPS}} = \text{Var}(\text{MMPS}_t) = \alpha' \Sigma \alpha = 1$.

The objective is to choose the linear combination of monetary policy surprises that maximizes the cumulative contribution of the MMPS_t to the forecast-error variance of a weighted average of target variables over horizons from 0 to H. Let w_i denote nonnegative weights such that $\sum_i w_i = 1$. Then the coefficients α of the MMP shock are determined by the following maximization problem:

$$\alpha^* = \arg\max_{\alpha} \sum_i w_i \sum_{h=0}^{H} \Omega_{i,\text{MMPS}}(h), \tag{10}$$

subject to the zero restrictions for the vector α and the restriction that the variance of the MMP shock is one (equations 3 and 4). As in Barsky and Sims (2011), the weights α can be recovered from principal components analysis as we discuss in Online Appendix A. This relationship to principal components analysis implies that additional monetary policy shocks—orthogonal to the MMP shock—can then be recovered as the second and, depending on k_M, subsequent principal components.[3]

3.3 Discussion

Before turning to its implementation and the results, we discuss the advantages and disadvantages of our empirical approach. An important way in which our estimation approach differs from the literature is that the MMP shock is not a structural shock that corresponds to a particular monetary policy action, but a reduced-form monetary policy shock that cap-

[3] Dou, Ho, and Lubik (2025) show that max-share identification is valid only under strict conditions, and that the identified shock can be contaminated when these conditions fail. This concern is unlikely to apply to our setting for two reasons. First, we do not try to identify a particular structural monetary policy shock, only the combination of them that best explains the target variables. Second, we build the MMP shock from the innovations to the monetary policy surprises, which, under our identifying assumption, cannot be contaminated by non-monetary shocks.

tures a mix of different policy actions. As the previous section makes clear, the procedure does not require the identification of structural shocks. Instead, the MMP shock is directly constructed from the reduced-form residuals of the monetary policy block. This is both an advantage and a limitation.

The upside is that the MMP shock captures the most important variation of monetary policy for a set of outcome variables in question (as determined by the weighting vector w) without the requirement that the researcher needs to take a stance on a particular identification scheme (or rotation). Many identification schemes have proven controversial in the literature. For instance, there is an ongoing debate as to whether information effects are important (Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Bauer and Swanson, 2023a).

The downside of identifying a reduced-form monetary policy shock is that it speaks less directly to the monetary transmission mechanism and that it is generally unclear how to map such a shock to existing models. For our main application, however, this limitation is immaterial. As long as the monetary policy surprises from which the MMP shock is constructed are the result of *some* monetary policy action—an assumption we maintain throughout—we can study the effects of this most important dimension of monetary policy on the macroeconomy and the explanatory power of the MMP shock is correctly attributed to monetary policy.

An added benefit of our procedure is that it avoids weak instrument problems that often arise when high-frequency identified monetary policy surprises are used as external instruments. Lewis and Mertens (2026) discuss the biases under weak instruments and how the estimates become biased and unreliable. To deal with this problem, Bauer and Swanson (2023b) add speeches of the Fed chair and other events to the list of FOMC announcements and show that this added variation greatly increases instrument strength. Adding more events, however, is not always possible, and we suspect that the lack of variation in monetary surprises is the chief reason why relatively few time series studies have analyzed the effects of multiple monetary policy shocks.

Our identification procedure can handle surprises with limited variation. It does not rely on the scaling inherent in instrumental variable estimation and related approaches that is at the heart of weak IV problems. If a monetary policy surprise with limited variation has small effects on the targeted macroeconomic variables, the procedure simply places a small loading on this monetary surprise. There are no implications for the consistency and reliability of the estimates (other than the loss of degrees of freedom that arises from the

additional inclusion of any variable in the VAR).

Again, however, this comes with a cost. While instruments only require a correlation with the instrumented variable, the monetary policy surprises included in the vector M_t need to be correctly measured. If the monetary policy surprises are subject to classical measurement error, the impulse response estimates are biased toward zero and their forecast error variance contributions understated (Plagborg-Møller and Wolf, 2022). Further, proxy VARs allow the researcher to estimate the VAR over a sample period that extends further back in time than the period for which high-frequency monetary policy surprises are available.

4 Implementation and Results

We next lay out a concrete implementation of our estimation procedure and then discuss the results.

4.1 VAR Specification

We study a standard monetary policy VAR as in Gertler and Karadi (2015), with two differences. First, we study the effects of three monetary policy surprises. Specifically, we use (i) a monetary policy surprise (MPS) about near-term rates similar to Bauer and Swanson (2023b), which is constructed from the first four Eurodollar futures contracts ED1-ED4, (ii) a surprise that captures variation in yields of longer maturities and is constructed from treasuries with 2-, 5-, 10-, and 30-year maturities (which we call PATH similar to Gürkaynak, Sack, and Swanson (2005) and Swanson (2021)) and (iii) the Boehm and Kroner (2024) non-yield shock. We orthogonalize all surprises with respect to the Bauer and Swanson (2023a,b) controls to prevent potential contamination from the Fed-response-to-news channel. The high-frequency changes of asset prices for the MPS and PATH surprises are obtained from the U.S. Monetary Policy Event-Study Database (USMPD) of Acosta, Ajello, Bauer, Loria, and Miranda-Agrippino (2025). Further details on the data are available in Online Appendix B. We note that the three monetary surprises are essentially uncorrelated with one another.

Second, we add additional variables to the VAR. The macro-finance block from Gertler and Karadi (2015) includes the log of industrial production, the log of the headline CPI, the 1-year Treasury rate, and the Gilchrist and Zakrajšek (2012) excess bond premium (EBP). To capture the dynamics of yields at longer maturities, we add the 5-year Treasury yield. We also add the log of the VIX and the log of the broad dollar index to capture variation in financial conditions and the strength of the dollar, which are both strongly affected by the non-yield shock (Boehm and Kroner, 2024).

We then estimate the VAR as discussed in Section 3. When doing so, we include $p = 12$ lags, and choose a weighting vector w that places a weight of 1/2 on industrial production and 1/2 on the headline CPI. In addition, we choose a target horizon for the objective of $H = 48$, implying that the MMP shock maximizes its cumulative variance contribution over four years. Our sample period runs from 1996m1 to 2025m7.[4] Standard errors are obtained using a block bootstrap.

4.2 Baseline Results

We now discuss the estimation results. We first illustrate the transmission of the NY shock, since its macroeconomic effects have not been explored. We then turn to our main findings on the MMP shock.

4.2.1 Effects of the Non-yield Shock

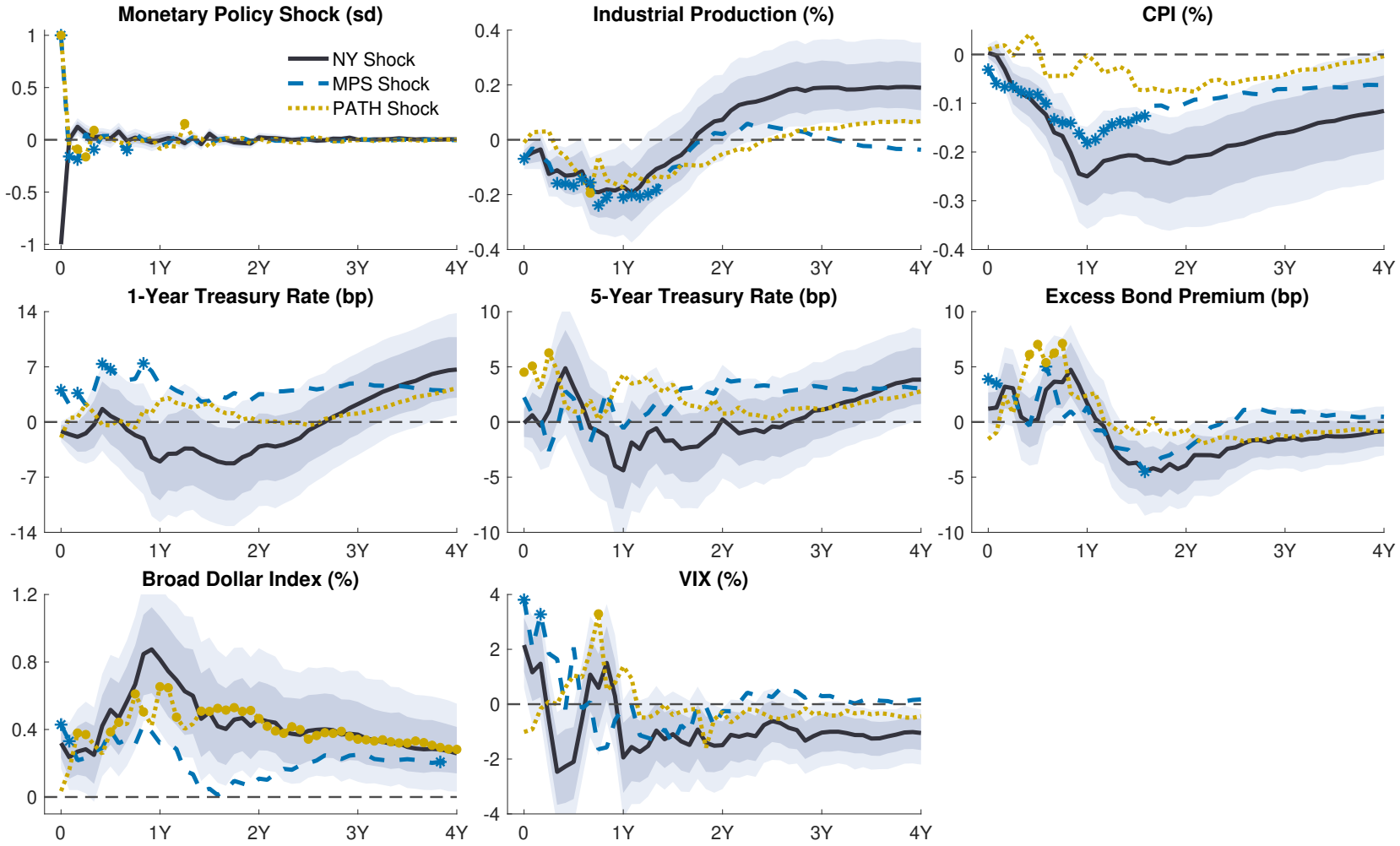
Figure 1 shows the impulse response functions to a one-standard-deviation contractionary NY shock and contrasts them with those to equally-sized contractionary MPS and PATH shocks. These impulse response functions are constructed with the shock in question ordered first in a recursive identification scheme. As the figure shows, the contractionary NY shock leads to a reduction in industrial production of about 0.2 percent with a trough after about one year and subsequent recovery. The CPI falls substantially and persistently by about 0.25 percent and remains significantly below zero for about three and a half years. This contrasts with the MPS shock, which has a similar effect on industrial production but a smaller effect on the CPI. The effects of the PATH shock have the expected signs, but are both smaller.

Since the NY shock is constructed to be orthogonal to interest rates at high frequencies, its impact effects on the 1- and 5-year Treasury rates are small and insignificant. Further, the effects remain small and insignificant over the full estimation horizon. As expected, the MPS shock raises the 1-year rate on impact, and the PATH shock raises the 5-year rate. Both rates remain elevated after the respective shock, though the estimates are noisy.

Turning to the effects on financial markets, both the MPS and the NY shocks raise the EBP on impact, appreciate the dollar, and raise the VIX. After about one year, the effects on the EBP and the VIX change signs (but are noisy), while the effect of the NY shock on the dollar grows and remains significant over the full estimation horizon. The PATH shock also has substantial effects on the dollar and a delayed effect on the EBP.

[4] For the episode from 2020m3 to 2020m7, we use surprises without orthogonalizing them with respect to the Bauer and Swanson (2023a,b) controls. The reason is that these controls take on extremely large values during this period, potentially doing more damage than good. In addition, the VAR includes five indicator variables for the months from 2020m3 to 2020m7 to "dummy out" the Covid episode, including the unorthogonalized surprises.

Figure 1: Impulse Response Functions to the NY, the MPS, and the PATH Shock



Notes: This figure plots the impulse responses of seven macro-financial variables to the non-yield shock (NY, solid black), monetary policy surprise (MPS, dashed blue), and path shock (dotted yellow), obtained from the VAR as specified in Section 4.1. The dark and light gray bands show the 68 and 90 percent confidence bands computed from a block bootstrap for the non-yield shock. Asterisks and circles indicate statistical significance of IRFs to MPS and PATH shocks, respectively, at the 90 percent confidence level.

Table 1: Explained Share of Forecast Error Variance (4-year average, %)

	Industrial Production	CPI	1Y Treasury Rate	5Y Treasury Rate	Excess Bond Premium	Broad Dollar Index	VIX
Panel A: High-Frequency Surprises							
MPS Shock	3.7	5.0	2.2	0.8	2.9	2.7	3.6
PATH Shock	2.0	0.7	0.3	1.6	4.8	5.8	1.5
NY Shock	5.0	14.0	1.1	0.9	4.9	10.6	4.4
Panel B: Max-share components							
MMP Shock	8.6	17.5	0.6	0.9	7.8	15.5	4.3
Second Shock	1.4	0.3	1.4	1.6	2.4	1.0	1.8
Third Shock	0.4	0.7	1.7	0.8	2.2	2.5	3.2

Notes: The table shows forecast error variance decompositions (FEVDs), averaged over four years from $h = 0$ to $h = 48$, for the different variables (columns) and the different shocks (rows). Panel A shows the FEVDs for the three monetary surprises included in M_t. Panel B shows the FEVDs for the Main, Second, and Third Monetary Policy Shocks, corresponding to the three principal components identified from the objective (A1) in Online Appendix A.

Panel A of Table 1 provides a forecast error variance decomposition for these three shocks and all variables in the macro-finance block. The numbers are point estimates of the explained share of forecast-error variance averaged over the horizons $h = 0, \cdots, 48$. Over the forecast horizon of four years, the MPS explains 3.7 percent of industrial production and 5.0 percent of the headline CPI. The PATH shock explains 5.8 percent of the broad dollar index and 4.8 percent of the EBP. The contributions of the NY shock are overall larger. It explains 5.0 percent of the forecast error variance of industrial production, a sizable 14.0 percent of the CPI, and 10.6 percent of the broad dollar index. In addition, it explains more variation of the EBP and the VIX than the MPS and PATH shocks. Clearly, the monetary policy shock variation captured by the NY shock is a non-negligible contributor to the U.S. business cycle.

4.2.2 The Main Monetary Policy Shock

The impulse responses of the MMP shock are shown in Figure 2. Since the three monetary policy shocks underlying the MMP shock are standardized and approximately orthogonal to one another, their impact responses are close to the loadings in the vector α^*. The non-yield shock receives the by far largest loading of about 0.8 (in absolute value), followed by the MPS and the PATH shock. Intuitively, the signs of the loadings are consistent with the contractionary directions shown in Figure 1. In the first one and a half years, a positive MMP shock resembles a negative demand shock in many ways: industrial production and the headline CPI decline, financial conditions tighten (the EBP and the VIX rise), and the dollar appreciates, while interest rates remain mostly unchanged. After that, we observe a

reversal in the effects on industrial production and financial conditions, while the price and dollar responses persist.

Figure 2 demonstrates that the MMP shock has large and persistent effects on the CPI and the dollar, and sizable effects on industrial production. This is also reflected in the forecast error variance contributions of the MMP shock, shown in Panel B of Table 1. When averaged over 48 months, the MMP shock explains about 9% of the variation in industrial production, about 18% in the headline CPI, and about 16% in the broad dollar index. This is our main substantive finding: U.S. monetary policy shocks do explain nontrivial shares of U.S. business-cycle fluctuations.

Panel B of Table 1 also reports the forecast error variance contributions of the second and third monetary policy shocks, that is, the second and third principal components identified from the objective (A1) in Online Appendix A. For all variables in the VAR except interest rates, the MMP shock accounts for the large majority of the variance explained by the three monetary policy shocks combined. For the 1- and 5-year Treasury rates—which are not among the target variables in the construction of the MMP shock—the second and third monetary policy shocks are more important.

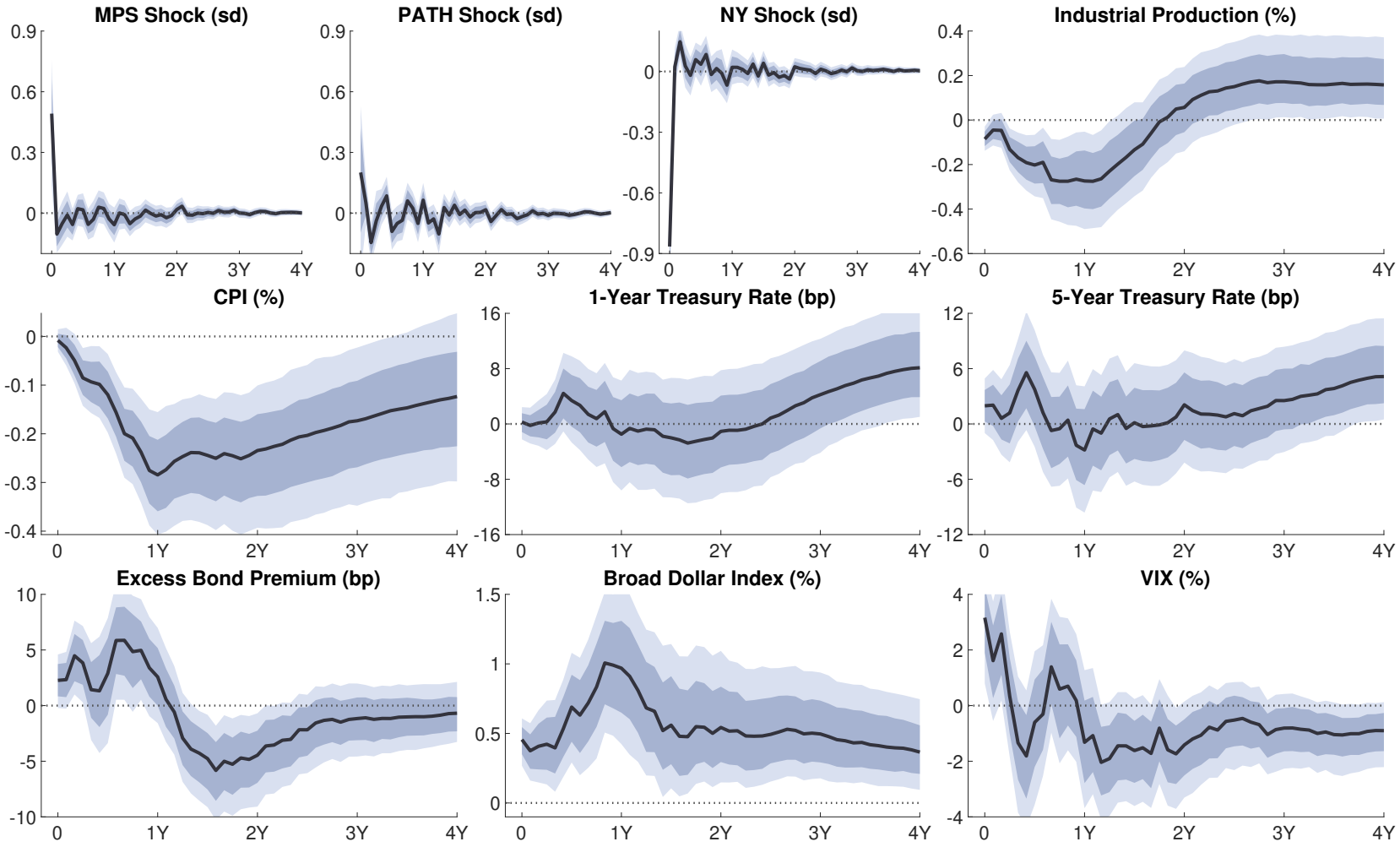
The MMP shock's large effects on macroeconomic variables warrant further investigation of its transmission. To do so, we run the following local projection specification:

$$y_{t+h} = c_h + \beta_h \text{MMPS}_t + \boldsymbol{\gamma}_h' \mathbf{X}_t + \nu_{t+h}, \qquad h = 0, 1, \ldots, 48, \tag{11}$$

where y_{t+h} is the (log of an) outcome variable h months after month t, β_h is the impulse response coefficient at horizon h, and $\mathbf{X}_t$ is a vector of control variables: 12 lags of the three monetary surprises, the macroeconomic and financial variables in the VAR, and the outcome variable of interest, as well as time dummies for the months 2020m3–2020m7. We consider six outcome variables: the core CPI, the Brent crude oil price, net exports to GDP, industrial production of durable consumer goods, the 30-year fixed mortgage rate, and housing starts.

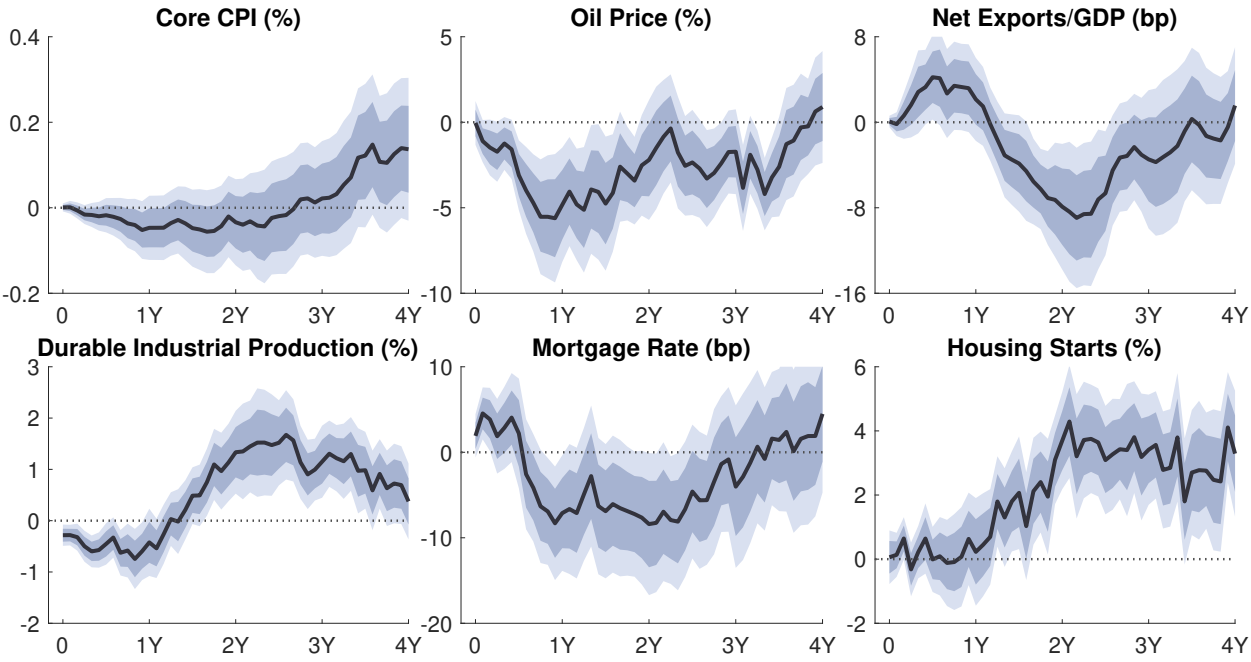
Figure 3 provides several insights into the transmission of the MMP shock. First, the large drop in oil prices, combined with a relatively muted decline in the core CPI, suggests that the effects of the MMP shock on the headline CPI are driven largely by energy prices. Second, the strong appreciation of the dollar is accompanied by a decline in net exports. Both the effect on oil prices and that on net exports suggest that the effect of the MMP shock is likely to spill over to other economies as well.

Figure 2: Impulse Response Functions to the Main Monetary Policy Shock



Notes: This figure plots the impulse response functions to the main monetary policy (MMP) shock, obtained from the VAR specified in Section 4.1. The black solid lines are point estimates, and the shaded areas show 68 and 90 percent confidence bands computed from a block bootstrap.

Figure 3: Additional Impulse Response Functions to the Main Monetary Policy Shock

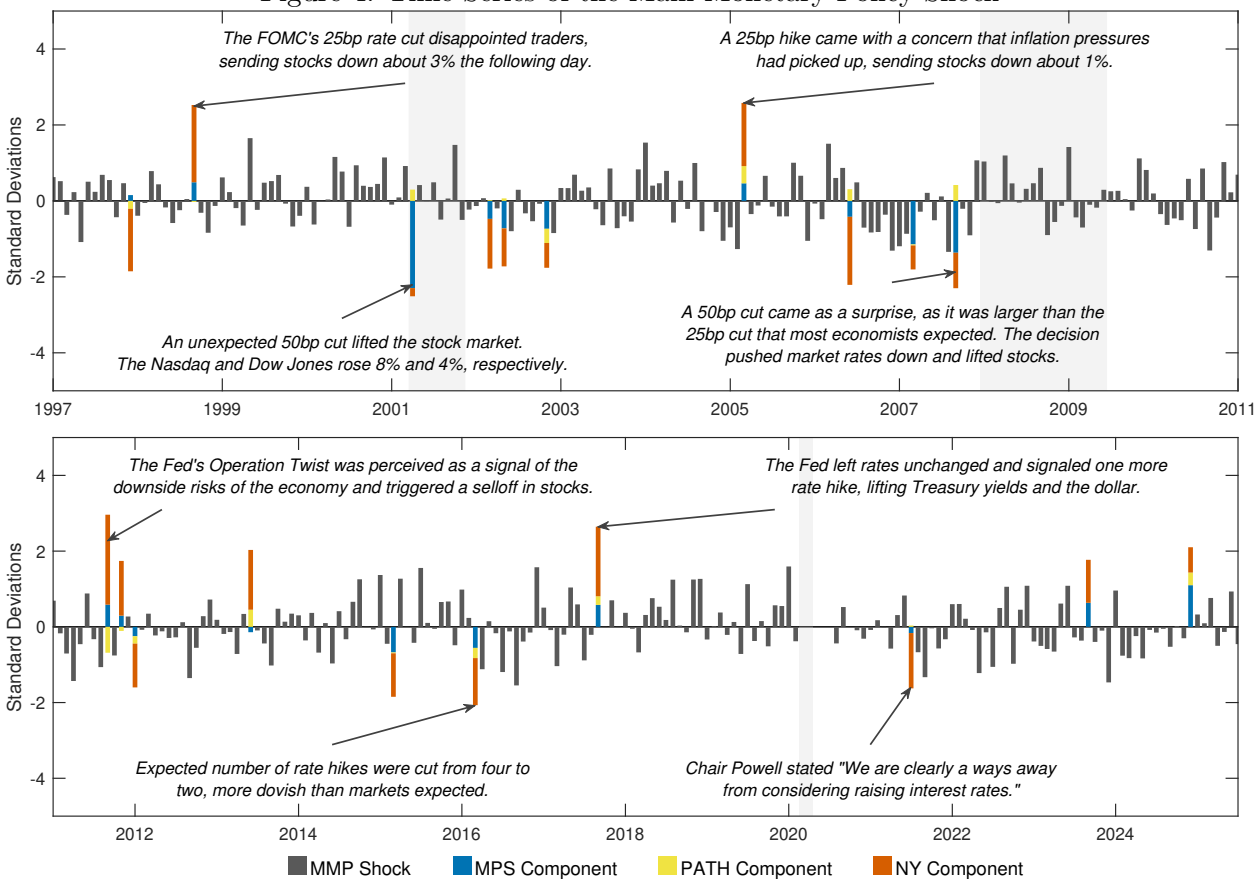


Notes: This figure plots the impulse response functions to the main monetary policy (MMP) shock, obtained using specification (11). The black solid lines are point estimates, and the shaded areas show 68 and 90 percent confidence bands computed from Newey-West standard errors.

Next, we return to a feature of the industrial production response from Figure 2. After about two years, the industrial production response to an initially contractionary MMP shock turns positive and significantly so after three years. The bottom three panels in Figure 3 explore possible reasons why this might be the case. Industrial production of durable goods initially falls but then recovers, suggesting that consumers and firms may replenish depleted stocks. In addition, mortgage rates fall about a year after a contractionary shock while housing starts begin to increase. Recalling from Figure 2 that the EBP also begins to fall after about a year, it thus appears that economic activity could also rebound due to an improvement in financial conditions.

Lastly, we inspect the time series of the MMP shock, shown in Figure 4 together with market commentary for some of the largest realizations in absolute value. The series is signed such that positive observations are contractionary and negative ones expansionary. The figure illustrates multiple points. First, the MMP shock exhibits substantial variation throughout the sample, rather than being concentrated in particular periods. Second, in all annotated cases, the market commentary is consistent with both the sign of the MMP shock and—intuitively—its composition. Third, the varying composition of the MMP shock

Figure 4: Time Series of the Main Monetary Policy Shock



Notes: This figure displays the time series of the Main Monetary Policy Shock. For a selected set of FOMC announcements, we show the decomposition into its MPS, PATH, and NY components, and text boxes with key market commentary. Grey bands indicate NBER recession periods.

realizations makes clear that on some dates the MPS shock is the dominant contributor and on other dates the NY shock. The PATH shock is typically not the dominant contributor. We provide more details in Online Appendix D.

4.3 Robustness and Extensions

We next provide a brief summary of several robustness checks and extensions. Details are relegated to Online Appendix C.

Different target horizons H The baseline MMP shock was identified over the target horizon $H = 48$. In Online Appendix C.1, we consider alternative target horizons $H \in \{12, 24, 36\}$. The MMP shock's variance contributions and impulse responses are very similar to the baseline when it is constructed for different target horizons.

8-variable VAR Since our sample is limited and our baseline VAR includes 10 variables, we estimate an alternative specification that drops the PATH shock and the 5-year rate (Online Appendix C.2). It turns out that the MMP shock, its impulse response functions, and its explanatory power are very similar to the baseline.[5]

Additional outcomes We study additional outcomes in Online Appendix C.3. There, we show that commodity prices fall more broadly (not only oil) and that prices of precious metals decrease after a contractionary MMP shock. In addition, survey measures of inflation expectations decline. Alternative measures of financial conditions confirm an initial deterioration and subsequent improvement. Sectoral employment responses are also consistent with the observed patterns of sectoral output and prices.

5 Conclusion

We estimate a main monetary policy shock as the linear combination of multiple high-frequency monetary policy surprises that best explains variation in industrial production and inflation. The estimation procedure does not require the identification of structural monetary policy shocks and is not susceptible to weak instrument problems. The resulting MMP shock efficiently summarizes the variation in U.S. monetary policy surprises that matters most for industrial production and consumer prices. Under the identification assumption that these surprises are not contemporaneously affected by non-monetary shocks, the MMP shock can be used to estimate causal effects.

Our findings not only question the view that monetary policy shocks are unimportant for business-cycle variation in recent decades, but also suggest that the monetary transmission mechanism differs in important ways from the conventional view in the literature. The inclusion of additional monetary policy surprises in the VAR, most notably the non-yield shock of Boehm and Kroner (2024), is critical for our findings. Together with the discussion in Boehm and Kroner (2024), our results for the MMP shock indicate that new identification schemes incorporating the variation captured by the non-yield shock are needed to understand the full scope of monetary policy transmission.

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[5] We also considered two four-shock variants: the three baseline surprises plus a fourth based on MP1 and MP2, which captures the shortest end of the yield curve, and the non-yield shock combined with the three shocks of Swanson (2021). In both cases, neither the explanatory power of the MMP shock nor its impulse response functions changed materially.

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Online Appendix
for
The Main Monetary Policy Shock*

November 3, 2026

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A Relationship to Principal Components Analysis

To see how the MMP shock can be recovered from principal components analysis, begin with constructing the lower-triangular Cholesky factor S_0 of the covariance matrix of reduced-form residuals, so that $\Sigma = S_0 S_0'$. Plugging this into the objective from equation (10) and rearranging gives

$$\sum_i w_i \sum_{h=0}^{H} \Omega_{i,\text{MMP}}(h) = \widetilde{\alpha}' \left(\sum_i w_i \widetilde{Q}_i(H) \right) \widetilde{\alpha},$$

where

$$\widetilde{Q}_i(H) = \sum_{h=0}^{H} \frac{S_0' \left(\sum_{\tau=0}^{h} \Phi_\tau' e_i e_i' \Phi_\tau \right) S_0}{e_i' \left(\sum_{\tau=0}^{h} \Phi_\tau \Sigma \Phi_\tau' \right) e_i}$$

and $\widetilde{\alpha} = S_0' \alpha$. This transformation of α converts the unit-variance normalization into a unit-length restriction, $\widetilde{\alpha}'\widetilde{\alpha} = 1$.

Next, let

$$\widetilde{Q}(H) = \sum_i w_i \widetilde{Q}_i(H).$$

Note that only the upper-left $k_M \times k_M$ block of $\widetilde{Q}(H)$, denoted by $\widetilde{Q}_{11}(H)$, enters the optimization, because the last k_X elements of the vector α are restricted to be zero. Defining $\widetilde{\alpha}_M$ as the first k_M elements of $\widetilde{\alpha}$ so that

$$\widetilde{\alpha}^* = \begin{pmatrix} \widetilde{\alpha}_M \\ 0_{k_X \times 1} \end{pmatrix},$$

the optimization problem therefore reduces to

$$\widetilde{\alpha}_M^* = \arg\max_{\widetilde{\alpha}_M} \widetilde{\alpha}_M' \widetilde{Q}_{11}(H) \widetilde{\alpha}_M \tag{A1}$$

subject to

$$\widetilde{\alpha}_M' \widetilde{\alpha}_M = 1.$$

Hence, the MMP shock can be constructed by first finding the largest eigenvalue of $\widetilde{Q}_{11}(H)$ with unit length, $\widetilde{\alpha}_M^*$, and then recovering the weights as $\alpha^* = (S_0')^{-1} \widetilde{\alpha}^*$.

Table B1: Data sources

Variable	Description	Source
Panel A: High-frequency monetary policy surprises		
ED1–ED4	Eurodollar Futures to construct the MPS	US Monetary Policy Database
Treasury yields	Maturities of 2-, 5-, 10-, and 30-year for the Path shock	US Monetary Policy Database
Non-yield shock	High-frequency shock orthogonal to yield changes	Boehm and Kroner (2024)
Panel B: Macroeconomic and financial variables in the VAR		
IP	Industrial production index	FRED: INDPRO
CPI	Consumer price index	FRED: CPIAUCSL
GS1	1-year Treasury constant-maturity yield (%)	FRED: DGS1
GS5	5-year Treasury constant-maturity yield (%)	FRED: DGS5
EBP	Excess bond premium (%)	Gilchrist and Zakrajšek (2012)
BDI	Broad U.S. dollar index: Nominal and Goods	FRB, H.10
VIX	CBOE Volatility Index	FRED: VIXCLS
Panel C: Outcome variables in the local projections		
Core CPI	CPI less food and energy	FRED: CPILFESL
Brent oil price	Brent crude oil spot price (US$ per barrel)	FRED: DCOILBRENTEU
Net exports / GDP	Nominal net exports of goods and services relative to nominal GDP (%)	FRED: NETEXP, GDP
Durable IP	Industrial production, durable consumer goods	FRED: IPDCONGD
Mortgage rate	30-year fixed-rate mortgage average (%)	FRED: MORTGAGE30US
Housing starts	New privately-owned housing units started	FRED: HOUST
Panel D: Additional outcome variables in the local projections		
S&P GSCI	Broad commodity price index	Bloomberg: SPGSCI
Gold price	Gold price (USD per troy ounce)	GoldPrice.com
Silver price	Silver price (USD per troy ounce)	GoldPrice.com
Michigan survey	Median expected price change over the next year (%)	FRED: MICH
Blue Chip	One-year-ahead CPI inflation forecast (%)	Blue Chip Economic Indicators
FCI-G	Financial Conditions Index	FRB, FCI-G
Baa yield	Moody's seasoned Baa corporate bond yield (%)	FRED: DBAA
Aaa yield	Moody's seasoned Aaa corporate bond yield (%)	FRED: DAAA
Total nonfarm	Total nonfarm sector employees	FRED: PAYEMS
Durable goods	Durable goods sector employees	FRED: DMANEMP
Construction	Construction sector employees	FRED: USCONS
Mining	mining, quarrying, and oil and gas extraction	FRED: CES1021000001

B Data Appendix

Table B1 lists the data used in the paper and their sources. The construction of the monetary policy surprises follows the literature. For the monetary policy surprise (MPS), we take the high-frequency changes in the first four quarterly Eurodollar futures (ED1–ED4) around FOMC announcements from the U.S. Monetary Policy Event-Study Database (USMPD) of Acosta et al. (2025) and obtain their first principal component. The shock is rescaled so that its impact on ED4 is unity.

For the path shock, we regress the high-frequency changes in the 2-, 5-, 10-, and 30-year Treasury yields from the USMPD on the MPS and take the first principal component of the residuals. For the non-yield shock, we use the Fed non-yield shock of Boehm and Kroner. We aggregate each surprise to monthly frequency by summing over the announcements within

a month, with months without an announcement set to zero. We then orthogonalize each series with respect to the six macroeconomic and financial controls of Bauer and Swanson (2023a,b) to purge out "the Fed response to news" channel. To address COVID outliers, we replace observations from 2020m3 to 2020m7 with the non-orthogonalized surprises as recommended by Bauer and Swanson (2023b) in the replication data description. But we dummy out those five months in our VAR estimation. Finally, each shock series is normalized to have a unit standard deviation.

In the VAR and local projections, for variables that are available at daily frequency, we use end-of-month values to avoid the time-aggregation problem. Lastly, the broad dollar index (BDI) is the nominal goods and services index from January 2006 onward. For the period before 2006, we extend the data back with the discontinued goods-only dollar index, rescaled so that the two series coincide on January 2, 2006. The two series have a correlation above 0.99 over the overlapping sample period (Jan 2, 2006 – Dec 31, 2019). Unless otherwise specified, the data span 1996m1 - 2025m7. Due to the initial 12 lags in the VAR, the estimation sample begins in 1997m1.

Table C1: Share of forecast error variance explained by the MMP shock, by target horizon (%)

	IP	CPI	GS1	GS5	EBP	BDI	VIX
Panel A: Average over horizons 0 to 48							
$H = 48$ (baseline)	8.6	17.5	0.6	0.9	7.8	15.5	4.3
$H = 36$	8.7	17.5	0.6	0.9	7.8	15.2	4.3
$H = 24$	8.6	17.3	0.6	0.9	7.7	14.7	4.3
$H = 12$	8.4	17.0	0.7	0.8	7.2	13.3	4.4
Panel B: Average over horizons 0 to H							
$H = 48$ (baseline)	8.6	17.5	0.6	0.9	7.8	15.5	4.3
$H = 36$	8.2	17.3	0.4	0.8	7.1	14.6	4.0
$H = 24$	7.8	14.9	0.5	0.8	5.5	12.9	3.6
$H = 12$	5.2	7.6	0.7	0.7	3.6	9.0	3.4

Notes: The table shows the share of the forecast error variance of each variable (columns) explained by the MMP shock identified with target horizon H in the max-share objective (rows). Panel A averages the shares over horizons 0 to 48 for every H. Panel B averages them over target horizons 0 to H. $H = 48$ is the baseline in Table 1.

C Robustness and Extensions

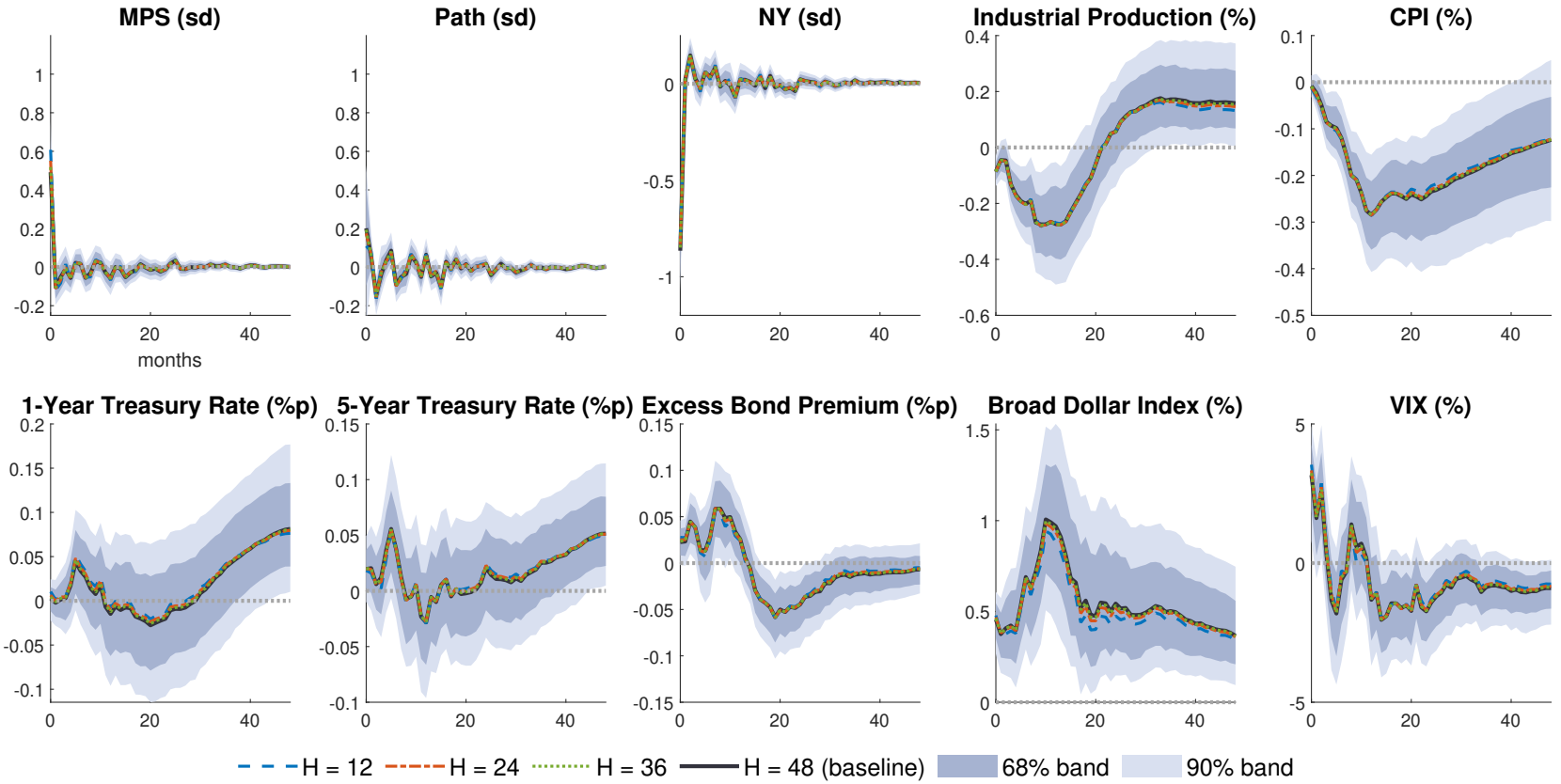
C.1 MMP shock identified over different horizons H

The baseline MMP shock is identified from equation (10) with the horizon set to $H = 48$. This section examines the robustness of the main results to the choice of target horizon by identifying the MMP shock over $H \in \{12, 24, 36\}$.

Table C1 compares the forecast-error variance decomposition results across the alternative target horizons. Panel A reports the contribution of each shock averaged uniformly over 48 horizons, and Panel B reports the contribution computed over each shock's own target horizon. The contributions in Panel A are similar across the choices of H. Panel B shows that the increase in the contribution over the target horizon is small beyond $H = 24$.

Figure C1 displays the impulse response functions of the macro-financial variables in the VAR to the MMP shocks with different target horizons. The point estimates for $H \in \{12, 24, 36\}$ are overlaid on the baseline impulse responses and bands for $H = 48$. The impulse responses under the alternative selections of H are very close to the baseline.

Figure C1: Impulse Response Functions to the MMP shock



Notes: This figure plots the impulse response functions to the main monetary policy shocks (MMP) that are identified over different target horizons H.

Table C2: Share of forecast error variance: Baseline vs. 8-variable VARs (%)

	IP	CPI	GS1	GS5	EBP	BDI	VIX
Panel A: Baseline VAR (10 variables)							
Monetary Policy Surprise	3.7	5.0	2.2	0.8	2.9	2.7	3.6
Path Shock	2.0	0.7	0.3	1.6	4.8	5.8	1.5
Non-Yield Shock	5.0	14.0	1.1	0.9	4.9	10.6	4.4
Main Monetary Policy Shock	8.6	17.5	0.6	0.9	7.8	15.5	4.3
Panel B: VAR without the path shock and GS5 (8 variables)							
Monetary Policy Surprise	3.3	7.3	1.9	–	2.3	1.9	4.3
Non-Yield Shock	6.0	14.3	1.1	–	6.3	9.4	4.4
Main Monetary Policy Shock	7.9	19.9	0.7	–	7.1	8.5	5.3

Notes: The table compares the forecast error variance decompositions of the baseline 10-variable VAR (Panel A, as in Table 1) with 8-variable VAR results that exclude the path shock and the GS5 (Panel B). The shares are averages over horizons 0 to 48.

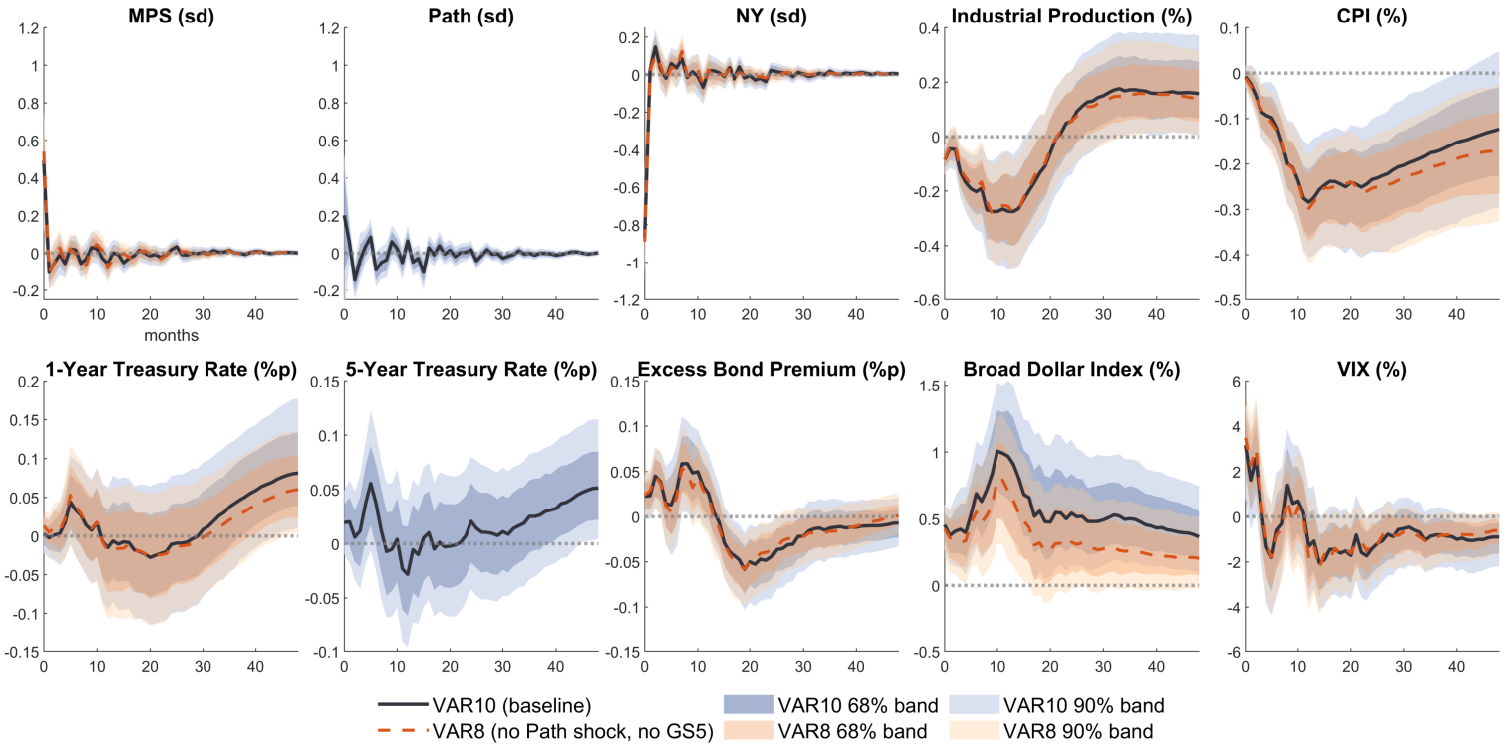
C.2 MMP shock identified with only MPS and Non-yield shock

Table C2 examines whether our results depend on the number of shocks used in the VAR. The MMP shock's explanatory power can be overstated if it depends on the number of shocks used in the VAR.

Panel B shows the estimation result without the path shock and the five-year Treasury yield. In this case, the MMP shock is the combination of the monetary policy surprise and the non-yield shock that maximizes the explained share of forecast error variance of IP and CPI. The results are close to the baseline. The MMP shock accounts for 7.9 % of the forecast error variance of IP and 19.9 % of that of CPI, compared with 8.6 and 17.5 % in the baseline. Its shares for the one-year Treasury yield, the excess bond premium, and the VIX are also similar.

Overall, the comparison shows that the explanatory power of the MMP shock for output and prices does not hinge on the number of shocks. Two shocks, the monetary policy surprise and the non-yield shock, are enough to recover an MMP shock with similar explanatory power.

Figure C2: Impulse Response Functions: Baseline vs. 8-variable VAR without Path shock



Notes: This figure plots and compares the impulse response functions to the main monetary policy shocks (MMP) identified from the baseline 10-variable VAR and the 8-variable VAR without the Path shock and the 5-year interest rate (GS5).

C.3 Additional Impulse Response Functions using Local Projections

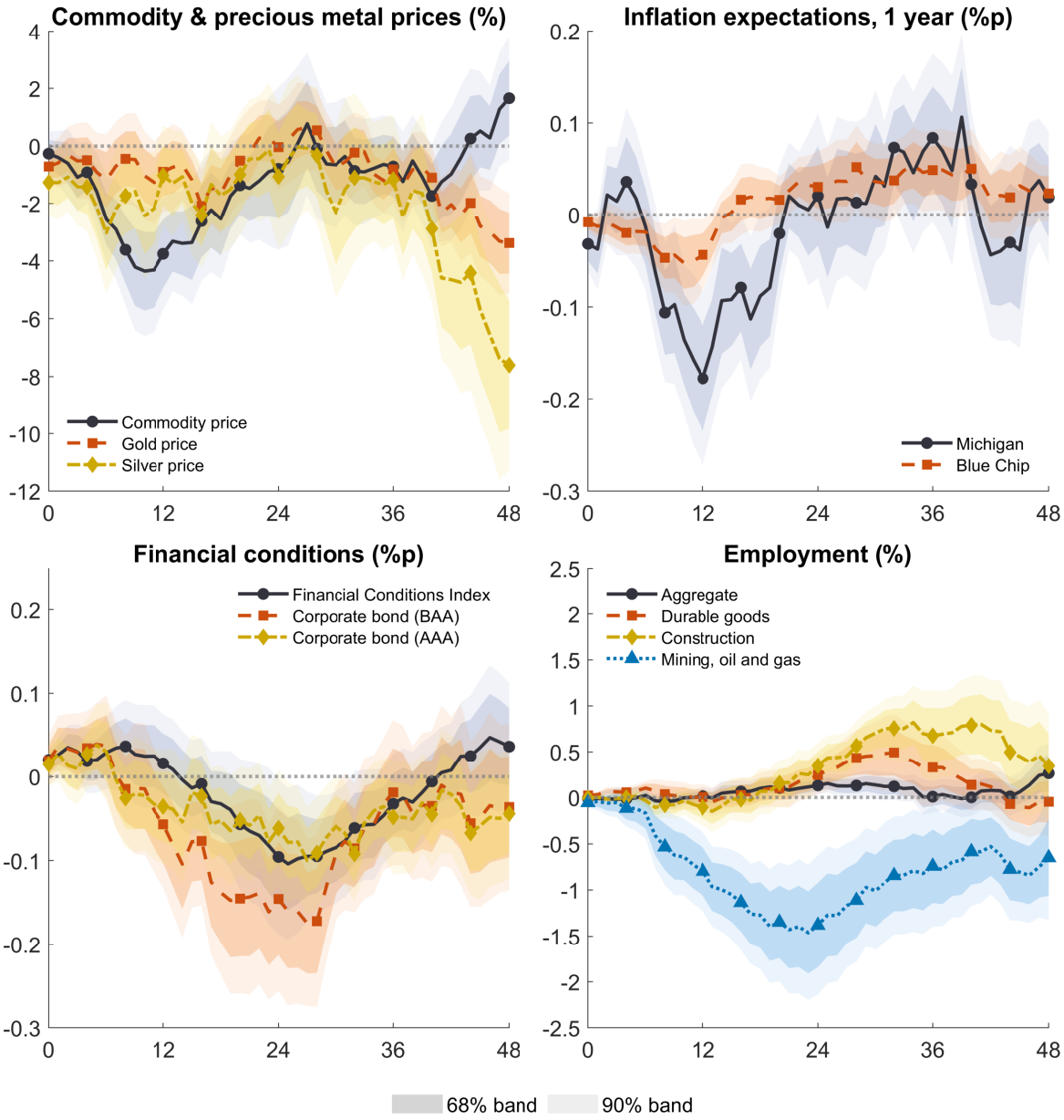
Figure C3 reports the responses of an additional set of variables to the MMP shock, estimated by local projections. The variables highlight four different dimensions of the MMP shock transmission. The first is commodity prices, measured by the S&P GSCI and the prices of precious metals (gold and silver). The second is inflation expectations, measured by one-year-ahead forecasts from the University of Michigan Surveys of Consumers and from Blue Chip Economic Indicators. The third is financial conditions, measured by the Federal Reserve Board's Financial Conditions Index (FCI-G) and Moody's Baa and Aaa corporate bond yields. The fourth is the labor market, measured by total nonfarm payroll employment and employment in durable goods, construction, and mining, quarrying, and oil and gas extraction. Commodity and precious metal prices and the employment series enter in their logs, multiplied by 100. The other variables enter in levels, and their responses are in percentage points.

The first two panels show the IRFs of the commodity price index, price of precious metals, and inflation expectations. As observed in the oil price, the MMP shock loads on these commodity prices. The contractionary response is especially strong for the commodity price index, with the trough observed around 4%. This is consistent with the baseline result that the MMP shock's contractionary effect on prices is mostly through the non-core component of the price index. The second panel, which shows the IRFs of inflation expectations from the survey data, also exhibits similar contractionary responses.

The third panel shows the responses of financial conditions, measured by the FCI and the two corporate bond yields. The responses mirror the response of the excess bond premium in the baseline VAR. Following a contractionary MMP shock, financial conditions tighten briefly, as the FCI and bond yields rise over the first six to twelve months. About two years after the shock, both bond yields and FCI decline in line with the pattern observed from the excess bond premium.

The last panel shows the responses of (log) employment. This aggregate employment shows a muted response, which masks considerable heterogeneity across sectors. Employment in mining, quarrying, and oil and gas extraction falls by about 1.5 percent, roughly two years after the shock. This is consistent with the fact that MMP shock loads especially heavily on commodity prices. In contrast, employment in durable goods manufacturing and construction exhibits a delayed expansion. This is consistent with the delayed boom in the industrial production of durable goods and in housing starts.

Figure C3: Additional Impulse Response Functions to the MMP shock using Local Projection



Notes: The figure shows impulse responses to a contractionary MMP shock, estimated by local projections over 1997m1–2025m7. For blue-chip inflation expectations, the estimation runs up to 2024m9 due to data subscription. Each regression includes a constant and 12 lags of the dependent variable and the set of variables in the baseline VAR specification. Observations whose shock date or outcome date falls in 2020m3–2020m7 are excluded. Shaded areas are 68% and 90% Newey-West standard error bands with 12 lags. The horizon is in months.

D Ex-post Narrative Account

Table D1: Details on Large Realizations of the Main Monetary Policy Shock

		Components			
Date (FOMC)	MMP	MPS	PATH	NY	Narrative
Panel A: 6 Largest Positive Realizations of MMP shocks (contractionary)					
09/20/2017	2.65	0.58	0.23	**1.84**	The Fed left rates unchanged and signaled one more rate hike, lifting Treasury yields and the dollar.
03/22/2005	2.58	0.46	0.45	**1.67**	A 25bp hike came with a concern that inflation pressures had picked up, sending stocks down about 1%.
09/29/1998	2.50	0.49	−0.03	**2.04**	The FOMC's 25bp rate cut disappointed traders, sending stocks down about 3% the following day.
09/21/2011	2.28	0.59	−0.68	**2.37**	The Operation Twist was perceived as a signal of the downside risks of the economy and triggered a selloff in stocks.
12/18/2024	2.10	**1.10**	0.33	0.67	A 25bp cut came with projections of only two cuts in 2025, pushing Treasury yields up and the Dow down more than 1,100 points.
06/19/2013	1.89	−0.14	0.45	**1.58**	Fed Chair Bernanke said asset purchases could slow later in 2013 and end by mid-2014, sending stocks and bonds lower and the 10-year yield to 2.33%.
Panel B: 6 Largest Negative Realizations of MMP shocks (expansionary)					
04/18/2001	−2.21	**−2.29**	0.30	−0.22	An unexpected 50bp cut lifted the stock market. The Nasdaq and Dow Jones rose 8% and 4%, respectively.
03/16/2016	−2.07	−0.56	−0.27	**−1.24**	Expected number of rate hikes was cut from four to two, more dovish than markets expected.
06/29/2006	−1.90	−0.42	0.31	**−1.79**	An expected 25bp hike came with language signaling a possible pause, sending the Dow up about 2% and the Nasdaq up 3%.
09/18/2007	−1.88	**−1.37**	0.42	−0.93	A 50bp cut came as a surprise and pushed market rates down and lifted stocks.
03/18/2015	−1.85	−0.68	−0.02	**−1.15**	The Fed dropped "patient" from its guidance but lowered its rate projections, lifting stocks more than 1% and weakening the dollar.
03/21/2007	−1.80	**−1.14**	−0.03	−0.63	The Fed held at 5.25% but dropped its reference to "additional firming" of rates, lowering the expected policy path and lifting the Dow 1.3%.

Notes: The table details the narrative analysis of FOMC announcements associated with large MMP shocks over the sample (1997m1–2025m7).